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MATHEMATICS

STANDARD FIVE

Monthly test - July 2024
Answer Key and Worked Solutions

Subject	Mathematics
Code	Not printed in source
Level	Standard Five
Year	2024
Exam Board / Issuer	HEAGTON
Exam	Monthly test - July
Type	Answer key and worked solutions
Questions	33 numbered questions/subparts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: Mathematics-S5-2024-HEAGTON-July-Monthly-test.pdf

Calculations

1(a). Add $37777 + 63333$.

Step 1: 1 place: $7 + 3 = 10$; write 0, carry 1.

Step 2: 10 place: $7 + 3 + \text{carry } 1 = 11$; write 1, carry 1.

Step 3: 100 place: $7 + 3 + \text{carry } 1 = 11$; write 1, carry 1.

Step 4: 1000 place: $7 + 3 + \text{carry } 1 = 11$; write 1, carry 1.

Step 5: 10000 place: $3 + 6 + \text{carry } 1 = 10$; write 0, carry 1.

Step 6: Write final carry 1 in the next place.

Step 7: $37777 + 63333 = 101110$.

Final Answer: 101110

1(b). Subtract $878984 - 878844$.

Step 1: 1 place: $4 - 4 = 0$.

Step 2: 10 place: $8 - 4 = 4$.

Step 3: 100 place: $9 - 8 = 1$.

Step 4: 1000 place: $8 - 8 = 0$.

Step 5: 10000 place: $7 - 7 = 0$.

Step 6: 100000 place: $8 - 8 = 0$.

Step 7: $878984 - 878844 = 140$; check $140 + 878844 = 878984$.

Final Answer: 140

1(c). Multiply 197×86 .

Step 1: Split into place values: $100 \times 86 = 8600$; $90 \times 86 = 7740$; $7 \times 86 = 602$.

Step 2: Add the partial products: 16942.

Final Answer: 16942

1(d). Divide 85000 by 5000.

Step 1: $8500 / 5000$: write 1; $1 \times 5000 = 5000$; $8500 - 5000 = 3500$.

Step 2: $35000 / 5000$: write 7; $7 \times 5000 = 35000$; $35000 - 35000 = 0$.

Step 3: $5000 \times 17 = 85000$.

Step 4: $85000 / 5000 = 17$.

Final Answer: 17

1(i). Subtract 999 from 9999.

Step 1: 1 place: $9 - 9 = 0$.

Step 2: 10 place: $9 - 9 = 0$.

Step 3: 100 place: $9 - 9 = 0$.

Step 4: 1000 place: $9 - 0 = 9$.

Step 5: $9999 - 999 = 9000$; check $9000 + 999 = 9999$.

Final Answer: 9000

1(j). Multiply 6894 by 16.

Step 1: Split into place values: $6000 \times 16 = 96000$; $800 \times 16 = 12800$; $90 \times 16 = 1440$; $4 \times 16 = 64$.

Step 2: Add the partial products: 110304.

Final Answer: 110304

3(b). 100 chalk sticks per box;154 boxes. Total?

Step 1: Split into place values: $100 \times 154 = 15400$.

Step 2: Add the partial products: 15400.

Final Answer: 15400 sticks

3(c). One pair of tongs costs 2354 shillings. Cost of 19 pairs?

Step 1: Split into place values: $2000 \times 19 = 38000$; $300 \times 19 = 5700$; $50 \times 19 = 950$; $4 \times 19 = 76$.

Step 2: Add the partial products: 44726.

Final Answer: 44726 shillings

6(a). Tree heights 355 cm, 455 cm, 204 cm. Sum?

Step 1: $355 + 455 = 810$.

Step 2: $355 + 455 + 204 = 1014$.

Step 3: $355 + 455 + 204 = 1014$.

Final Answer: 1014 cm

6(b). A phone rings 142 times daily for 1 week. Total?

Step 1: Split into place values: $100 \times 7 = 700$; $40 \times 7 = 280$; $2 \times 7 = 14$.

Step 2: Add the partial products: 994.

Final Answer: 994 times

6(c). Juma walks 231 steps daily for 45 days. Total?

Step 1: Split into place values: $200 \times 45 = 9000$; $30 \times 45 = 1350$; $1 \times 45 = 45$.

Step 2: Add the partial products: 10395.

Final Answer: 10395 steps

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Mathematics

1(e). Calculate $\frac{2}{7} + \frac{4}{14}$.

Step 1: Common denominator 14: $\frac{2}{7} = \frac{4}{14}$.

Step 2: $\frac{4}{14} + \frac{4}{14} = \frac{8}{14}$.

Step 3: $\frac{8}{14} = \frac{4}{7}$.

Final Answer: $\frac{4}{7}$

1(f). Calculate $9/14 - 8/28$.

Step 1: Common denominator 28: $9/14 = 18/28$.

Step 2: $18 - 8 = 10$.

Step 3: $10/28 = 5/14$.

Final Answer: $5/14$

1(g). Multiply $5/6 \times 6/12$.

Step 1: Cancel 6 in numerator and denominator.

Step 2: $5/12$ remains.

Final Answer: $5/12$

1(h). Find $72 \times 5/6$.

Step 1: $72 / 6 = 12$.

Step 2: $12 \times 5 = 60$.

Final Answer: 60

2(a). Form largest number from 1,5,9,2,6 each once.

Step 1: Arrange greatest first: 9,6,5,2,1.

Final Answer: 96521

2(b). Write 611 pupils in Roman numerals.

Step 1: $600 = DC$; $11 = XI$.

Final Answer: DCXI

2(c). Expand 53042.

Step 1: Read all five place values.

Final Answer: $50000 + 3000 + 0 + 40 + 2$

2(d). Find the smallest digit value in 245034.

Step 1: Values 200000,40000,5000,0,30,4.

Step 2: 0 is smallest.

Final Answer: 0

2(e). Write 1308 in words.

Step 1: $1308=1000+300+8$.

Final Answer: One thousand three hundred and eight

2(f). Find the LCM of 42 and 18.

Step 1: $42=2 \times 3 \times 7$; $18=2 \times 3^2$.

Step 2: Use greatest prime powers: $2 \times 3^2 \times 7 =126$.

Final Answer: 126

3(a). Continue 168,129,90,51,[].

Step 1: Subtract 39 each step.

Step 2: $51 - 39 = 12$.

Final Answer: 12

4(a). List primes in 79 through 95.

Step 1: Reject even numbers and multiples of 3,5,7.

Step 2: 79,83,89 have no divisor up to their square roots.

Final Answer: 79,83,89

4(b). Find difference between largest and least odd numbers in 79 through 95.

Step 1: Greatest 95, least 79.

Step 2: $95 - 79 = 16$.

Final Answer: 16

4(c). Write smallest even number from 79 through 95 in Roman numerals.

Step 1: First even number 80.

Step 2: $80 = \text{LXXX}$.

Final Answer: LXXX

5(a). Difference between CMXLIV and DCLXXIX?

Step 1: $\text{CMXLIV} = 900 + 40 + 4 = 944$.

Step 2: $\text{DCLXXIX} = 500 + 100 + 50 + 20 + 9 = 679$.

Step 3: $944 - 679 = 265$.

Final Answer: 265 (CCLXV)

5(b). Find the GCF of 42 and 18.

Step 1: $42 = 2 \times 3 \times 7$; $18 = 2 \times 3^2$.

Step 2: Shared primes 2×3 .

Step 3: $2 \times 3 = 6$.

Final Answer: 6

5(c). List factors of 45.

Step 1: Factor pairs $1 \times 45, 3 \times 15, 5 \times 9$.

Final Answer: 1,3,5,9,15,45

7(a). Complete factor tree 51 with branches 3 and[].

Step 1: $51 \div 3 = 17$.

Final Answer: 17

7(b). Factor tree 30 splits 2 and 15;15 splits 3 and[].

Step 1: $15 \div 3 = 5$.

Step 2: $30 = 2 \times 3 \times 5$.

Final Answer: 5

8(a). Pentagon has 5 equal sides 21 cm. Find perimeter.

Step 1: $5 \times 21 = 105$.

Final Answer: 105 cm

8(b). Isosceles triangle sides 8 cm, 8 cm, 9 cm. Perimeter?

Step 1: $8 + 8 + 9 = 25$.

Final Answer: 25 cm

8(c). Coffee pictogram: July 5 triangles, October 9 triangles, each 108 kg. Difference?

Step 1: $9 - 5 = 4$ symbols.

Step 2: $4 \times 108 = 432$.

Final Answer: 432 kg

Section Summary: Read each value and unit carefully. Align place values and check calculations by reversing the operation. For pictures, count only the requested objects.

Source Checks

Darasa Huru - HEAGTON Standard Five (V) Monthly Test July 2024